Programme Name	B. Tech. (Computer Engineering)	**Semester**	VII
Course Code	R5CO4004L		
Course Title	**Cross Platform App Development Lab**		
Course Scheme	(L-T-P: 0-0-2) (ISCE : 60-40)		
Credits	1		
Course Type	VSEC		
Prerequisite: Java, Operating System			
COURSE OUTCOMES: Students will be able to **CO1.** Develop applications using Flutter components and widgets. **CO2.** Develop applications using Dart programming components. **CO3.** Develop cross-platform applications using React Native and its components. **CO4.** Develop applications using Xamarin components and layouts.			

Cross Platform App Development Lab Experiments

	COURSE CONTENTS	Hrs	CO
1.	Develop applications using Flutter widgets.	2	1
2.	Develop different layout structures using Row, Column, and Stack widgets in Flutter.	2	1
3.	Develop adaptive and responsive designs in Flutter.	2	1
4.	Develop a map application using Dart.	2	2
5.	Develop an HTML DOM application using Dart.	2	2
6.	Develop cross-platform applications using React Native.	2	3
7.	Develop an expense manager application using widgets in React Native.	2	3
8.	Develop a connectivity-based cross-platform application using React Native.	2	3
9.	Develop Xamarin applications consisting of widgets.	2	4
10.	Develop Xamarin applications using layouts and views.	2	4

TEXT BOOKS	
1.	Michael Katz, Kevin Moore, Vincent Ngo, *Flutter Apprentice: Learn to Build Cross-Platform Apps*, 2012, ISBN: 1950325113.
2.	Akshat Paul, Abhishek Nalwaya, *React Native for Mobile Development*, Apress Publication, 2019, ISBN: 978-1484244531.
3.	Dan Hermes, *Xamarin Mobile Application Development: Cross-Platform C# and Xamarin.Forms Fundamentals*, Apress, 2015, ISBN: 978-1484202142.
RECOMMENDED BOOKS	
1.	Thomas Bailey, Alessandro Biessek, *Flutter for Beginners*, Packt Publishing, ISBN: 978-1800565999.